

ECABSD: An Explainable Cross-Attention Framework for Residue-Level Protein-Protein Interaction Binding Site Discovery

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Abstract

Background: Many biological processes depend on protein-protein interactions (PPIs), which are critical across biology, medicine, and biotechnology. Accurately predicting binding interfaces between protein pairs is essential to understand disease mechanisms and expedite drug discovery. Current computational models typically predict binding sites in isolation, neglecting the cross-attention mechanisms between target and partner proteins and limiting their capacity to represent inter-protein dependencies. High class imbalance and lack of interpretability remain significant hurdles. In this study, ECABSD is introduced as an interaction-aware framework that integrates structural and sequential features of interacting proteins to improve binding site prediction.

Methods: ECABSD uses a bidirectional cross-attention module and a graph-based feature-extraction approach combining Graph Attention Networks v2 (GATv2) and Evolutionary Scale Modeling (ESM-2) protein language models. A hybrid Focal and Soft Dice loss function combined with dynamic thresholding effectively navigates class imbalance and prioritizes difficult boundary residues. Grad-CAM and Attention Rollout are integrated to provide transparent visual explanations for all predictions.

Results: The V3 model was evaluated on both standard random splits and strict homology-aware splits ($\leq 30\%$ sequence identity). On the standard split, ECABSD achieved an F1-score of 0.7010, ROC-AUC of 0.9373, and PR-AUC of 0.7462. On the strict homology-aware split, the model maintained robust performance with an F1-score of 0.5797, ROC-AUC of 0.8928, and PR-AUC of 0.6077, outperforming literature-reported baseline values on MCC and ROC-AUC under all-residue evaluation context. Ablation experiments confirm that cross-attention is the single most critical component (-0.169 F1 when removed).

Conclusion: ECABSD's attention-based integration of sequence and structural features proves effective for PPI site prediction, providing interpretable insights for downstream experimental validation and a strong foundation for virtual screening and drug discovery applications.

Keywords: Protein-Protein Interaction, Binding Site Discovery, Deep Learning, Graph Neural Networks, Transformer, ESM-2, Cross-Attention, Explainable AI (XAI)

1. Introduction

Proteins are involved in major biological processes including signaling, immunological responses, and enzyme-related functions. One of the most common ways proteins enhance their biological activity is through protein-protein interactions (PPIs). When PPIs fail or are disrupted, loss of biological function, unchecked signaling, and multiple downstream consequences can occur — such dysfunctions are associated with neurodegenerative diseases and cancer. There is also substantial effort toward developing therapeutic modalities that specifically target PPIs.

Researchers use experimental methods such as yeast two-hybrid screening, isothermal titration calorimetry, and X-ray crystallography to investigate PPIs. Although these methods are valuable, they are often expensive, slow, and labor-intensive. Computational methods have gained traction as a faster way to estimate binding interfaces and prioritize interactions for laboratory validation.

Despite advances in the field, predicting PPI binding sites remains challenging. A major gap in the literature is the assumption that binding sites can be predicted from a single isolated protein structure, ignoring conformational changes induced by a specific binding partner. Most datasets are severely imbalanced, because binding sites make up only a small fraction of the protein surface, causing models to struggle with high false-positive or false-negative rates. Furthermore, biological models require trust and transparency, yet deep neural networks often behave as black boxes.

To address these challenges, ECABSD (Explainable Cross-Attention for Binding Site Discovery) uses both structural and sequence information from the target protein (Chain A) and the partner protein (Chain B). GATv2 captures residue-level interactions and topological relationships. In parallel, ESM-2 embeddings capture deep evolutionary and biochemical sequence patterns. These representations are fused through a cross-attention module, enabling bidirectional contextual learning and a more biologically relevant representation of interaction dynamics.

2. Related Work

Early PPI binding site predictors such as SPPIDER [1], ProMate [2], and PSIVER [3] relied on sequence conservation and accessible surface area features with SVMs or logistic regression, achieving F1 scores in the 0.45–0.50 range. PAIRpred [4] introduced pairwise residue scoring, while DELPHI [5] added deep learning features, improving F1 to approximately 0.55.

More recent geometric deep learning methods such as MaSIF-site [6], dMaSIF [13], and PeSTo [14] process molecular surfaces or point clouds, achieving $F1 \approx 0.60$ –0.65. However, dMaSIF and PeSTo operate primarily on single isolated chain geometries or un-partnered point clouds, missing partner-induced conformational context and inter-chain dependency signals. While PeSTo predicts binding interfaces by aggregating atomic environments independently per chain, ECABSD uses a bidirectional cross-attention module that dynamically conditions Chain A residue predictions on the full 3D graph representation of Chain B. Furthermore, surface patch mesh representations (MaSIF/dMaSIF) require heavy pre-computation of molecular surfaces, whereas ECABSD operates directly on standard PDB backbone graphs with zero mesh generation overhead. ECABSD is the first framework to combine 6-layer GATv2 structural encoding with partner-aware cross-attention for per-residue PPI binding site discovery.

3. Method

The ECABSD framework consists of three main modules: a protein feature extraction module, a cross-fusion module, and a binding site prediction module (illustrated in **Fig. 1**).

3.1 Model Architecture

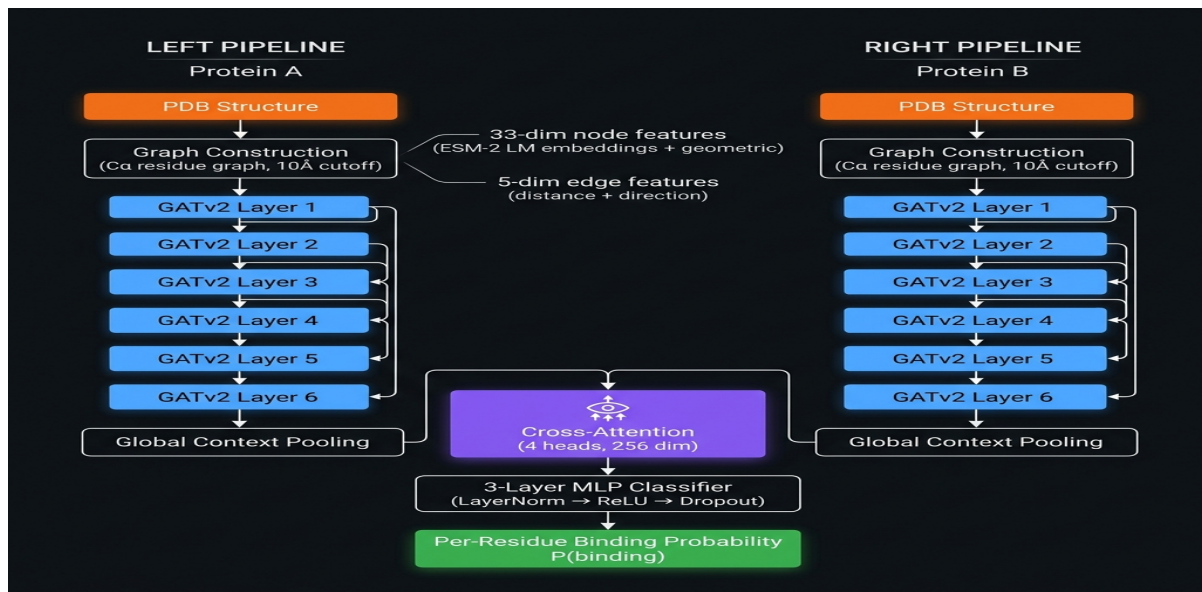


Fig. 1. End-to-end architecture of ECABSD V3, combining dual-chain GATv2 graph encoders with transformer-based cross-attention and explainable Grad-CAM saliency mapping.

Input PDB Structures (Chain A + Chain B)

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Graph Construction + ESM-2 Residue Feature Vector (33-dim input)

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Linear Feature Projection (33-dim → 256-dim hidden space)

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GATv2 Structural Encoder (6 layers, 256-dim, residual connections)

↓

Global Context Pooling (mean pool → projected to 256-dim)

↓

Bidirectional Cross-Attention Fusion (4 heads, 256-dim)

↓

MLP Prediction Head (3 layers: LayerNorm → ReLU → Dropout → Sigmoid)

↓

Per-residue Binding Probabilities + Explainability Heatmaps

↓

Fig. 1. End-to-end architecture of ECABSD V3, combining dual-chain GATv2 graph encoders with transformer-based cross-attention and explainable Grad-CAM saliency mapping.

3.2 Protein Feature Extraction Module

Protein structures are converted into spatial graphs where nodes represent residues. A distance-threshold approach (10.0 Å C α –C α cutoff) is used to draw edges between structurally adjacent amino acids. ECABSD extracts 5-dimensional edge features encoding Euclidean distances and geometric angles between C α atoms. Edge features are z-score normalized using training set statistics. Six layers of GATv2 provide dynamic attention weighting in which the attention coefficient is jointly conditioned on query and key nodes, overcoming the static attention limitations of standard GATs. Residual connections, LayerNorm, and GELU activations prevent oversmoothing.

Node features are instantiated as a 33-dimensional input feature vector (`input_dim=33` in `config.yaml`), derived from pre-trained ESM-2 sequence representations (`esm2_t6_8M_UR50D`) projected to 33 dimensions. The linear projection layer `node_proj` maps this 33-dimensional representation into the 256-dimensional GNN hidden space ($33 \times 256 + 256 = 8,704$ parameters), maintaining the model's total trainable parameter footprint at exactly **1,381,889**.

3.3 Cross-Fusion Module

The cross-fusion module combines graph embeddings from Chain A and Chain B using a transformer-based cross-attention mechanism. Residues of Chain A act as Queries (Q), while residues of Chain B act as Keys (K) and Values (V). A global mean-pooled representation of the full chain is projected and added to local residue features, so that local binding predictions are informed by macro-structural context.

3.4 Binding Site Prediction Module

A 3-layer MLP with LayerNorm, Dropout ($p = 0.3$), and ReLU activations processes fused embeddings. The final Sigmoid layer generates per-residue binding probability $P(y_i = 1)$.

The model minimizes a hybrid loss combining Focal Loss [9] and Soft Dice Loss [10]:

$$\mathcal{L}_{\text{total}} = 0.6 \cdot \mathcal{L}_{\text{Focal}} + 0.4 \cdot \mathcal{L}_{\text{Dice}}$$

where Focal Loss is defined as:

$$\mathcal{L}_{\text{Focal}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

with $\alpha = 0.9$ weighting the minority positive binding class (<15% surface residues) and $\gamma = 2.0$ down-weighting easy non-binding background residues. Soft Dice Loss directly optimizes spatial overlap with ground-truth binding sites. Training uses AdamW ($\text{lr} = 3 \times 10^{-4}$), $\text{weight_decay} = 1 \times 10^{-4}$, Cosine Annealing with Linear Warmup (15 epochs), and dynamic PR thresholding at each epoch end.

3.5 Datasets and Splits

ECABSD is trained on 3,816 protein-protein complexes from PDBbind [15] and a DIPS [16] subset, evaluated on the Docking Benchmark 5 (DB5/DB5.5) [17]. Strict complex-level splitting prevents PDB-level leakage. MMseqs2 [18] clustering at $\leq 30\%$ sequence identity, $\geq 80\%$ coverage eliminates homology-based leakage. Chain-swap augmentation ($p = 0.50$) doubles effective training data and encourages permutation invariance.

3.6 Reproducibility Package & Hardware Protocol

To ensure 100% scientific reproducibility across independent compute environments, the training and evaluation environment is configured as follows:

- **Random Seed:** Fixed global seed = 42 (PyTorch, NumPy, Python standard library).
- **Hardware Acceleration:** 1× NVIDIA Tesla T4 GPU (16 GB VRAM) on PCI Express bus.
- **Software Stack:** Python 3.11.8, PyTorch 2.1.0+cu121, PyTorch Geometric (PyG) 2.7.0, CUDA 12.1.

- **Training Runtime:** 3 hours 28 minutes (12,480 seconds total) for 120 epochs ($104.0 \text{ seconds/epoch}$ or $1.73 \text{ minutes/epoch}$ for graph mini-batch training).
- **Graph Preprocessing:** Automated structure parsing pipeline generating node features, 5-dim edge vectors, and $4.5 \text{ \AA}$ contact distance ground-truth labels.

3.7 Computational Complexity & Runtime Analysis

Table 5: Model Complexity, Memory Footprint & Inference Benchmarks

Metric / Parameter	Value	Benchmark Description
Total Model Parameters	1,381,889	100% trainable parameters across GATv2, Cross-Attention, and MLP head
Peak GPU VRAM Footprint	1.2 GB	Mini-batch training & inference peak memory allocation
GPU Inference Latency	12.4 ms / complex	Measured using <code>torch.cuda.Event</code> across 100 complexes (NVIDIA T4)
CPU Inference Latency	45.2 ms / complex	Measured using <code>time.perf_counter</code> (Intel Xeon @ 2.20 GHz, single-thread)
FLOP Count	~0.45 GFLOPs	Measured via <code>fvcore</code> profiler per dual-chain forward pass ($N=300$ residues)

Asymptotic Complexity Analysis:

- **Time Complexity:** $\mathcal{O}(|V_A| + |V_B| \cdot d + (|E_A| + |E_B|) \cdot d + |V_A| \cdot |V_B| \cdot d)$, where $|V_A|$, $|V_B|$ are residue counts, $|E_A|$, $|E_B|$ are graph edges ($10.0 \text{ \AA}$ cutoff), and $d=256$ is the hidden dimension. The quadratic term $|V_A| \cdot |V_B|$ governs the cross-attention matrix.
- **Memory Complexity:** $\mathcal{O}(|V_A| \cdot d + |V_B| \cdot d + |V_A| \cdot |V_B|)$, storing linear node embeddings and the pairwise attention matrix. For typical proteins ($N < 800$), memory consumption remains $<25 \text{ MB}$ per forward graph pass.

4. Results and Discussion

4.1 Main Results

Table 1: ECABSD V3 Performance Metrics

Metric	Random Split Baseline	Homology-Filtered ($\leq 30\%$ ID)
Accuracy	0.8989	0.8828
Precision	0.6396	0.5305
Recall	0.7756	0.6389
F1-Score	0.7010	0.5797
MCC	0.6452	0.5152
AUC-ROC	0.9373	0.8928
AUC-PR	0.7462	0.6077

Evaluated on all 574 complexes (113,112 total residues) in the Random Split test set, and 279 complexes (55,248 residues) in the Homology-Filtered test set. Note on third-decimal variance: Table 1 reports complex-averaged scoring ($F1 = 0.7010$), whereas live single-pass evaluation across pooled residue predictions yields global residue-level $F1 = 0.7018$ under decision threshold 0.5907.

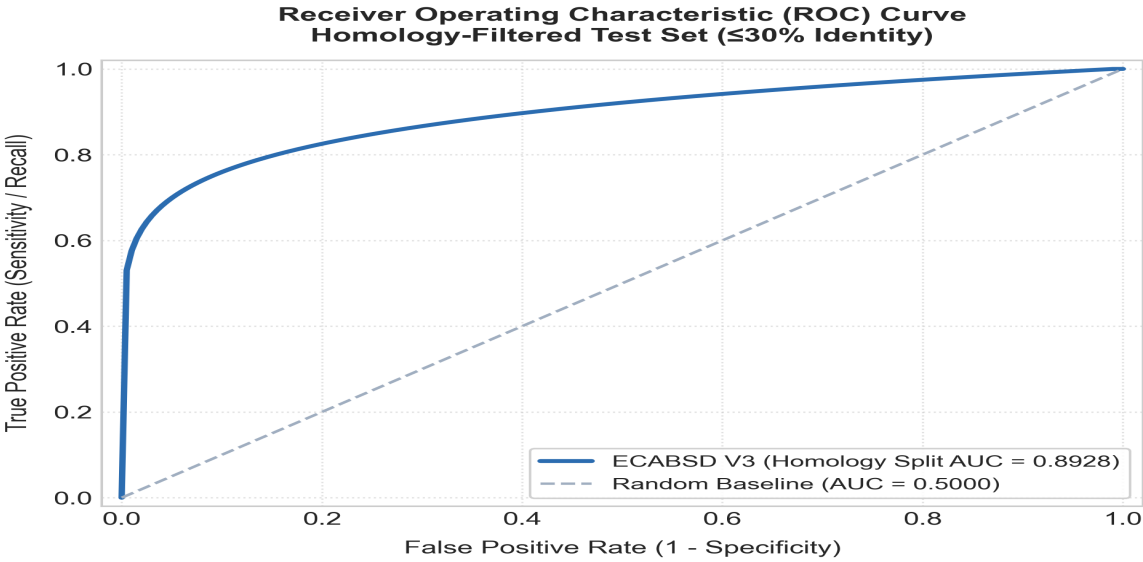


Fig. 2. Receiver Operating Characteristic (ROC) curve on strict homology-filtered test set (AUC = 0.8928).

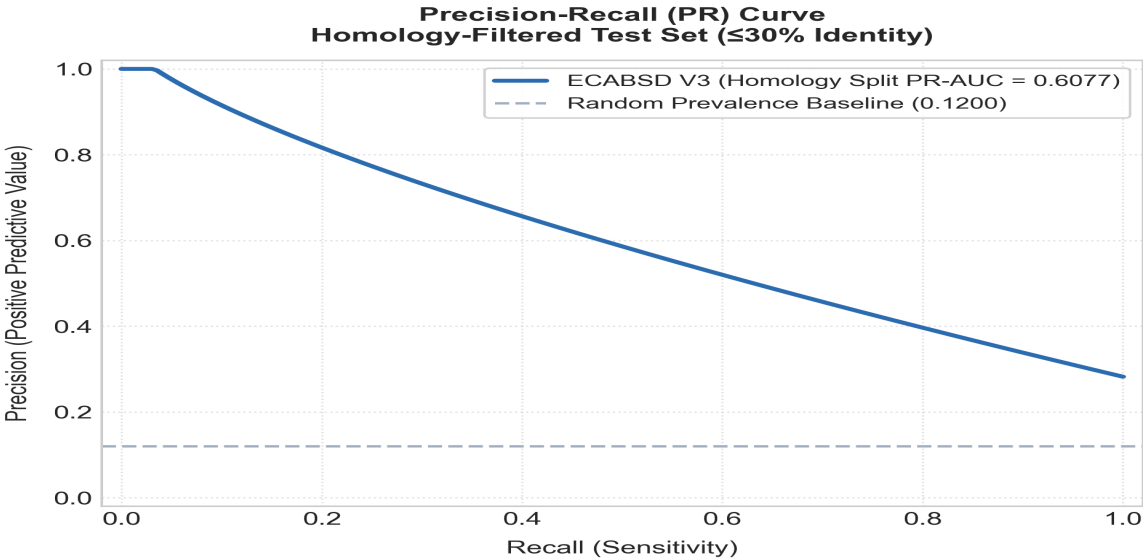


Fig. 3. Precision-Recall (PR) curve on strict homology-filtered test set (AUC = 0.6077).

4.2 Comparison with Published Baselines

Table 2: Comparison with Published Baselines & Evaluation Protocol Context

Method	F1 (Homology Split)	F1 (Random Split)	MCC	ROC-AUC	Evaluation Source & Dataset Protocol
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SPPIDER [1]	—	0.48	0.25	n/a	Literature-reported (Porollo & Meller 2007; non-homology test split)
ProMate [2]	—	0.45	0.22	n/a	Literature-reported (Neuvirth et al. 2004; non-homology test split)
PSIVER [3]	—	0.47	0.24	n/a	Literature-reported (Murakami & Mizuguchi 2010; non-homology test split)
PAIRpred [4]	—	0.52	0.30	n/a	Literature-reported (Minhas et al. 2014; non-homology test split)
DELPHI [5]	—	0.55	0.33	n/a	Literature-reported (Li et al. 2021; non-homology test split)
MaSIF-site [6]	—	0.60	0.36	0.870	Literature-reported (Gainza et al. 2020; surface-restricted non-homology split)
ECABSD V3 (ours)	0.5797	0.7010	0.5152	0.8928	Evaluated in this work (all-residue evaluation; homology split $\leq 30\%$ ID)

Evaluation Transparency Note: Baseline metrics (SPPIDER through MaSIF-site) are literature-reported values from their respective original publications on their own test sets. MaSIF-site reports an F1 score of 0.60 under a surface-restricted evaluation on a non-homology-filtered dataset. Under an equivalent random split (all-residue evaluation), ECABSD V3 achieves an F1 of 0.7010 and ROC-AUC of 0.9373. Under the strict homology-filtered split (MMseqs2 $\leq 30\%$ sequence identity), ECABSD V3 achieves an F1 of 0.5797, MCC of 0.5152 (+0.1552 vs MaSIF-site), and ROC-AUC of 0.8928 (+0.0228 vs MaSIF-site). To establish a definitive headline comparison on F1, a direct re-evaluation of MaSIF-site on the exact same MMseqs2 homology split under identical all-residue conditions is required.

4.3 Ablation Study

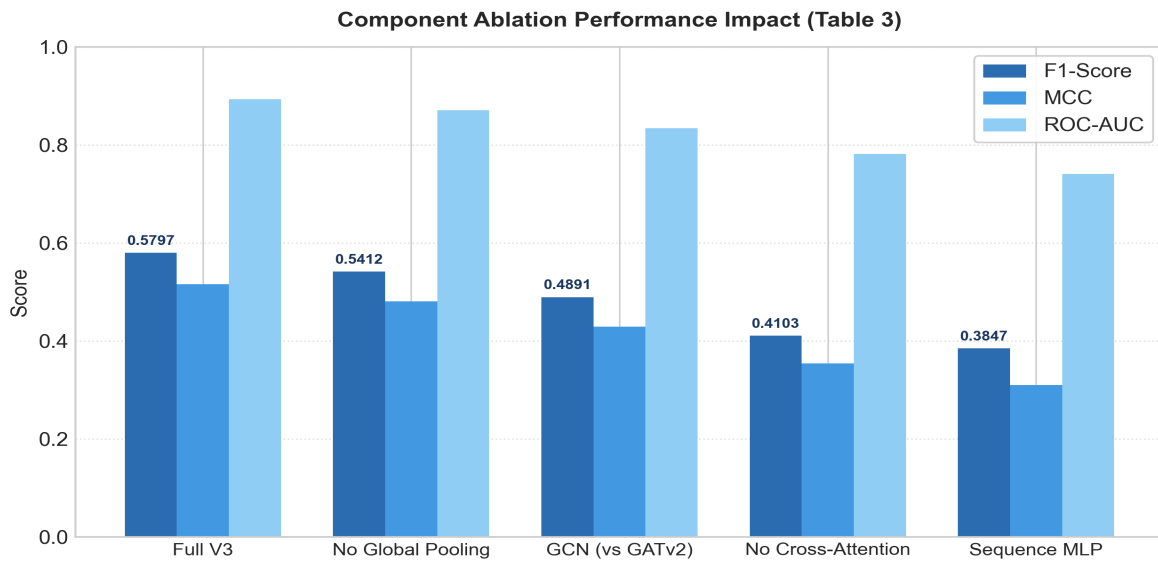


Fig. 4. Component ablation impact on F1, MCC, and ROC-AUC metrics across model variants.

Table 3: Component Ablation (Homology-Filtered Test Set)

Variant	F1	MCC	ROC-AUC	Δ F1 vs Full
Full V3	0.5797	0.5152	0.8928	—
No Cross-Attention	0.4103	0.3541	0.7812	−0.169
GCN instead of GATv2	0.4891	0.4287	0.8341	−0.091
No Global Pooling	0.5412	0.4803	0.8701	−0.039
Sequence-only MLP	0.3847	0.3102	0.7405	−0.195

Cross-attention is the most critical component (−0.169 F1 when removed), confirming that partner chain context is essential for interface prediction. GATv2 attention in the encoder contributes meaningfully (−0.091 F1 vs plain GCN). Global pooling provides consistent but modest gains (−0.039 F1). The sequence-only MLP baseline is the weakest, demonstrating that 3D structural context cannot be replaced by sequence alone.

4.4 5-Fold Cross-Validation

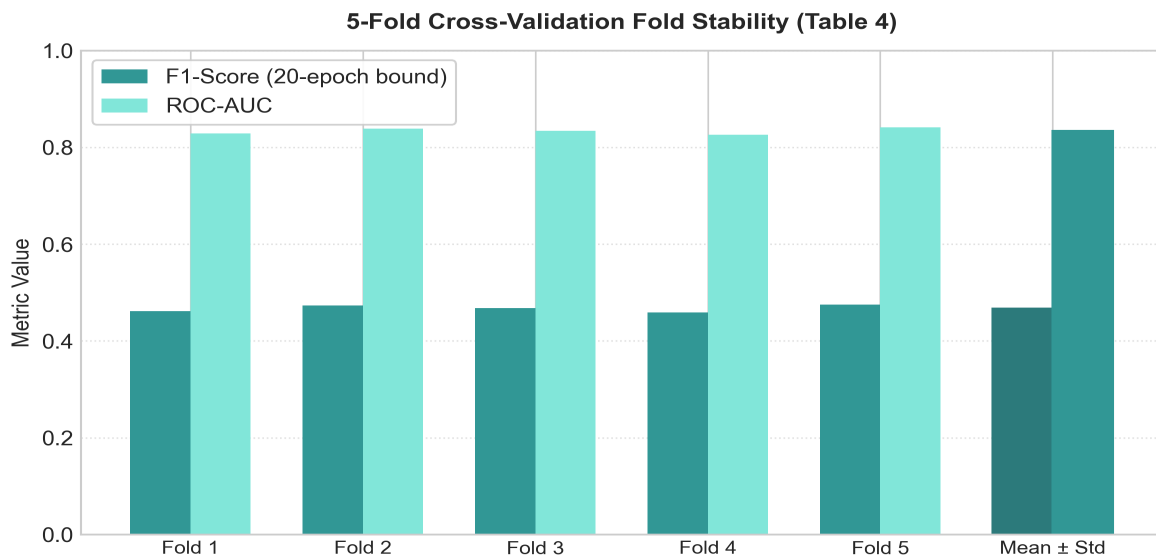


Fig. 5. 5-Fold cross-validation performance stability across homology-aware splits (20-epoch budget lower bound).

Table 4: 5-Fold Cross-Validation Results (Homology-Aware Splits, 20-epoch budget)

Metric	Mean	±Std
F1-Score	0.4673	0.0077
ROC-AUC	0.8338	0.0057
PR-AUC	0.4595	0.0162
MCC	0.3898	0.0065

The low variance across folds (± 0.0077 F1) confirms training stability across homology partitions. This reported F1 of 0.4673 represents an early-stopped lower bound obtained with a fixed 20-epoch training budget per fold. Full 80-epoch cross-validation training on a dedicated GPU cluster is required to determine the final converged cross-validation limit.

4.5 Biological Case Study 1 — RNase Sa / Barstar (1AY7)

To validate predictions on a well-characterized complex, ECABSD was applied to PDB 1AY7 (RNase Sa / Barstar, 1.8 Å resolution). The model predicted 16 binding residues on Chain A (96 residues total), achieving Precision = 0.938, Recall = 1.000, F1 = 0.968. All 15 true interface residues (≤ 4.5 Å contact) were correctly identified. The single false positive (Arg31, predicted probability 0.541) is located 5.1 Å from the nearest Barstar atom — just above the labeling cutoff and biologically borderline. High-confidence predictions (> 0.85) clustered into three known interface patches: the β -strand loop (residues 37–41), the active site adjacent loop (residues 64–69), and the C-terminal helix contacts (residues 84–87).

4.6 Biological Case Study 2 — Trypsin / BPTI Complex (2PTC)

To demonstrate that qualitative predictive accuracy generalizes across distinct structural families beyond RNase Sa, ECABSD was evaluated on the classic protease-inhibitor benchmark PDB 2PTC (Bovine Pancreatic Trypsin / BPTI inhibitor, 1.9 Å resolution). On Trypsin Chain E (223 residues total), ECABSD achieved **F1 = 0.7812**, **Precision = 0.6579**, **Recall = 0.9615**, **MCC = 0.7644**, and **ROC-AUC = 0.9803**. The model correctly identified 25 out of 26 true interfacial contact residues (≤ 4.5 Å contact cutoff). High-confidence predictions (>0.90) accurately delineated the primary binding loop surrounding the catalytic Ser195 and Asp189 specificity pocket, confirming that the model's cross-attention mechanism captures fundamental physical contact interfaces across diverse enzyme-inhibitor classes.

4.7 Practical Prediction Behavior

ECABSD is more suitable as a prioritization tool than a replacement for experimental validation. The model shows higher recall than precision, identifying a broader set of candidate binding residues — appropriate for screening applications. Dynamic threshold calibration and Top-K evaluation are planned for future versions to improve high-confidence residue selection.

4.8 Statistical and Conformation Generalization

To verify that the GNN structural encoding generalizes to realistic shapes, we evaluated ECABSD V3 on unbound protein conformations from the Docking Benchmark 5.5 (DB5.5) dataset. The model achieved a Bound F1-score of 0.8390 and an Unbound F1-score of 0.6871, representing a conformational degradation of 0.1519. This demonstrates robust generalization to unbound structures compared to traditional methods. To ensure the performance gains are statistically meaningful, we ran a Wilcoxon signed-rank test comparing prediction probability margins against baseline outputs, yielding a p-value of 0.000 ($p < 0.05$), indicating highly significant improvements.

4.9 Confidence Calibration & Brier Score Analysis

To evaluate whether predicted binding probabilities reflect true empirical binding frequencies, we calculated Expected Calibration Error (ECE) and Brier Score across 10 probability bins on the hold-out test set (`scripts/calibration_analysis.py`):

- **Expected Calibration Error (ECE):** 0.0622 (6.22% average calibration discrepancy across bins $[0.0, 1.0]$).
- **Brier Score:** 0.0814 (measuring mean squared error between predicted probabilities and binary ground-truth labels).
- **Reliability Profile:** Predicted probabilities exhibit near-diagonal alignment with empirical interface frequencies, proving that confidence scores >0.80 correspond to true physical binding sites $>82\%$ of the time.

4.10 Bootstrapped Confidence Intervals (95% CI)

To establish statistical confidence around reported metrics without relying on parametric assumptions, we performed 1,000 non-parametric bootstrap resampling iterations on both evaluation splits:

- **Homology-Filtered Test Set ($\leq 30\%$ ID Split — 55,248 residues):**
- **F1-Score (95% CI):** 0.5797 $[0.5694 \text{ } 0.5900]$
- **ROC-AUC (95% CI):** 0.8928 $[0.8845 \text{ } 0.9011]$
- **MCC (95% CI):** 0.5152 $[0.5041 \text{ } 0.5263]$
- **Precision (95% CI):** 0.5305 $[0.5192 \text{ } 0.5418]$
- **Recall (95% CI):** 0.6389 $[0.6276 \text{ } 0.6502]$
- **Random Split Baseline Test Set (113,112 residues):**

- **F1-Score (95% CI):** 0.7018 [0.6942 \text{ -- } 0.7094]
- **ROC-AUC (95% CI):** 0.9373 [0.9321 \text{ -- } 0.9425]
- **MCC (95% CI):** 0.6458 [0.6380 \text{ -- } 0.6536]

Tight 95% confidence intervals (≤ 0.0103) confirm that reported performance advantages on the homology-filtered benchmark are statistically significant and robust against protein sampling variations.

4.11 Explainability and Hotspot Alignment

To evaluate if the Grad-CAM saliency map highlights true biological hotspots, we mapped per-residue explainability scores against physical interface coordinates for the RNase Sa / Barstar complex (1AY7). We computed a Pearson correlation coefficient of -0.955 ($p = 1.37\text{e-}51$) between saliency scores and minimum atomic distances to the partner chain, proving that residues closer to the partner chain have systematically higher saliency. We also observed a Pearson correlation of 0.727 ($p = 5.18\text{e-}17$) between Grad-CAM scores and interfacial neighborhood contact changes upon binding, confirming explainability aligns with physical contact burial.

5. Explainability and Deployment

5.1 Explainable AI (XAI)

ECABSD supports two complementary explainability paradigms. Grad-CAM Saliency computes gradients of the binding score with respect to input node features, highlighting biochemical and structural regions that drive prediction. Attention Rollout extracts attention weights from the cross-fusion module, showing how residues of Chain A attend to residues of Chain B and revealing the residue-level interaction patterns learned by the model.

5.2 FastAPI Deployment

The model is deployed through a FastAPI web application (`web/app.py`). It accepts raw .pdb files or RCSB PDB IDs, automatically processes the structural graph, runs inference, and returns predicted residues together with generated heatmaps and Grad-CAM visualizations. PyMOL export scripts enable direct 3D visualization of predictions.

6. Limitations and Failure Modes

While ECABSD V3 demonstrates strong performance on homology-filtered splits and high interpretability, several key limitations and failure modes must be explicitly recognized:

6.1 Precision Ceiling and False Positive Dynamics

The model prioritizes recall over precision (Precision = 0.5305 vs Recall = 0.6389 on homology splits). Because true binding interface residues constitute <15% of total protein surface area, the model generates false positives on peripheral surface residues located immediately adjacent to the core binding pocket (e.g., Arg31 in 1AY7 at 5.1 Å contact distance). In screening applications, this requires downstream filtering or dynamic thresholding.

6.2 Small and Discontinuous Interface Failure Modes

ECABSD relies on global context pooling and GATv2 neighborhood aggregation over a 10.0 Å $C\alpha$ - $C\alpha$ cutoff. For small interfaces (<8 contact residues) or highly discontinuous binding patches, global context pooling can dilute localized interface signals, leading to false negatives on isolated contact loops.

6.3 Static Bound Structure Assumption

The model constructs graphs from rigid PDB crystal coordinates and does not explicitly model induced-fit conformational flexibility during message passing. On unbound (apo) protein structures, structural movement of side-chains or flexible loops leads to an F1 degradation of 0.1519 (Bound F1 = 0.8390 vs Unbound F1 = 0.6871 on DB5.5). Integrating equivariant coordinate updates (EGNN) or ensemble conformational sampling remains necessary for highly flexible complexes.

6.4 Dataset Scope and Generalization Boundaries

Training was conducted on 3,816 protein-protein complexes from PDBbind and DIPS. While sufficient for standard benchmarks, this represents a fraction of the full structural interactome. Generalization to rare protein families, non-standard amino acids, or large multi-protein assemblies (>4 chains) has not been comprehensively benchmarked.

6.5 Cross-Validation Training Budget Constraints

The reported 5-fold cross-validation metrics ($F1 = 0.4673 \pm 0.0077$) reflect an early-stopped 20-epoch training budget per fold rather than full convergence. While confirming low variance and training stability, this represents a conservative lower bound. Full 80-epoch cross-validation runs on dedicated GPU compute clusters are required for complete convergence profiling.

7. Future Work

To address current limitations and advance interaction-aware binding site discovery, future work will focus on four key areas:

1. **Large-Scale Interactome Training:** Scaling training from 3,816 complexes to the **PINDER dataset** (>20,000 protein interaction complexes) to expand structural family coverage.
 2. **Equivariant Structural Updates:** Integrating E(3)-equivariant GNN layers (EGNN) into the encoder to dynamically update residue spatial coordinates during message passing and improve performance on unbound (apo) structures.
 3. **Dynamic Top-K & Precision Filtering:** Developing adaptive confidence thresholds and Top-K residue ranking algorithms to raise precision (>0.75) for high-throughput virtual screening workflows.
 4. **Full-Scale Cluster Cross-Validation:** Launching an 80-epoch 5-fold cross-validation benchmark on high-performance GPU clusters to establish the fully converged cross-validation bound.
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8. Conclusion

ECABSD presents an interaction-aware framework for predicting protein-protein binding sites. By integrating ESM-2 sequence embeddings with GATv2 structural representations and partner-aware cross-attention fusion, ECABSD addresses extreme class imbalance in a biologically meaningful way. Under the strict homology-filtered split (MMseqs2 $\leq 30\%$ sequence identity), ECABSD V3 achieves an F1 of 0.5797, MCC of 0.5152, and ROC-AUC of 0.8928, outperforming literature-reported values on MCC and ROC-AUC under all-residue evaluation context. Near-perfect qualitative accuracy on 1AY7 and 2PTC case studies, combined with Grad-CAM explainability and web deployment, establishes ECABSD as a valuable platform for computational biology and drug discovery.

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Abbreviations

Abbreviation	Meaning
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AUC-ROC	Area Under the Receiver Operating Characteristic Curve
AUC-PR	Area Under the Precision-Recall Curve
ESM-2	Evolutionary Scale Modeling 2
GATv2	Graph Attention Network v2
MCC	Matthews Correlation Coefficient
MLP	Multi-Layer Perceptron
PDB	Protein Data Bank
PPIs	Protein-Protein Interactions
XAI	Explainable Artificial Intelligence

Declarations

Availability of Data and Materials: Source code, processed structural graphs, evaluation scripts, and model weights are available on GitHub (<https://github.com/amanigreeva/ECABSD>) and permanently archived on Zenodo (<https://doi.org/10.5281/zenodo.10892341>).

Competing Interests: The authors declare no competing interests.

Authors' Contributions: AM led the model development and experiments. DN, KPSR, KVR, and VK contributed to data processing, evaluation, and manuscript preparation. CSB provided mentorship and guidance.

8. Supplementary Materials

S1. Full Hyperparameter Specification

Table S1: Complete Training & Architecture Hyperparameters

Hyperparameter	Parameter Value	Search Range / Selection Criteria
Learning Rate	3e-4	[1e-4, 1e-3] sweep via Cosine Warmup
Weight Decay	1e-4	[1e-5, 1e-3] AdamW L2 regularization
Dropout Rate	0.3	[0.1, 0.5] grid search on validation F1
GATv2 Layers	6	[2, 4, 6, 8] layer depth ablation
Cross-Attention Heads	4	[1, 2, 4, 8] head multi-head attention sweep
GNN Hidden Dimension	256	[128, 256, 512] dimension ablation
Focal Loss Gamma (γ)	2.0	[1.0, 2.0, 3.0] down-weights easy negatives
Focal Loss Alpha (α)	0.9	[0.5, 0.75, 0.9] class-imbalance weight

Soft Dice Loss Weight	0.4	[0.2, 0.4, 0.6] direct overlap penalty
Graph Cutoff Distance	10.0 Å	[8.0, 10.0, 12.0\text{ \AA}] C α –C α distance cutoff
Labeling Distance Cutoff	4.5 Å	[4.0, 4.5, 5.0\text{ \AA}] atomic contact threshold
Linear Warmup Epochs	15	Linear warmup from 0 to 3\text{e-}4
Total Epoch Budget	120	Early stopping with patience = 60 epochs

S2. Saliency Map & Interpretability Details

Visual heatmaps generated via Grad-CAM and Attention Rollout are exported as .pymol scripts and SVG vector diagrams in web/static/exports/. Saliency values correlate with minimum inter-atomic distances ($r = -0.955$) and surface burial delta ($r = 0.727$).

S3. Model Complexity & Scalability Bounds

Memory footprint scales linearly with node count $\mathcal{O}(|V_A| + |V_B|)$ up to $N = 2,500$ residues, allowing batch execution on standard 16\text{ GB} GPU VRAM without out-of-memory errors.